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CSCI 5561 - Computer Vision
November 14, 2026

HW3 Report

Assignment item	Points	Location in report	Extra credit?
Report Quality	10	-	No
Gather Photographs	10	Section 1, p. 2	No
Estimate Camera Parameters	10	Section 2, p. 3	No
Two-Image 3D Reconstruction with SVD	15	Section 3, p. 6	No
Multi-Image Reconstruction with COLMAP	15	Section 4, p. 9	No
Gaussian Splatting	15	Section 5, p. 11	No
Triangulation Refinement	10	Section 7.1, p. 13	No
Learned Matching	10	Section 7.2, p. 13	No
Held-Out View Evaluation	10	Section 7.3, p. 13	Yes
Total attempted before extra-credit conversion	110		
Maximum possible score after conversion	105		

Cover Page: Feature and Rubric Accounting Table

Introduction

[Introduction text to be added here]

1 Gather Photographs

The scene here is the center table in the living room of my apartment. I placed multiple objects on the table with different colors, shapes and depths to get a good reconstruction My stratergy was to capture the table from all 4 sides, cover wide range of angles. I captured photos from the top view as well as some close to the objects.

Iphone 17 main camera used. 40 photos captured initially but deleted two due to motion blur. Total images now are 38. Original resolution of photos was 4284x5712. I shared it over Whatsapp to myself which automatically downsampled the images to 1200x1600. I am using image_27.jpg and image_29.jpg for two-view reconstructions. The camera in image_27.jpg sits in front of the camera in image_29.jpg. There is a lot of feature overlap and also covers the objects in scene from different angles, which is useful.

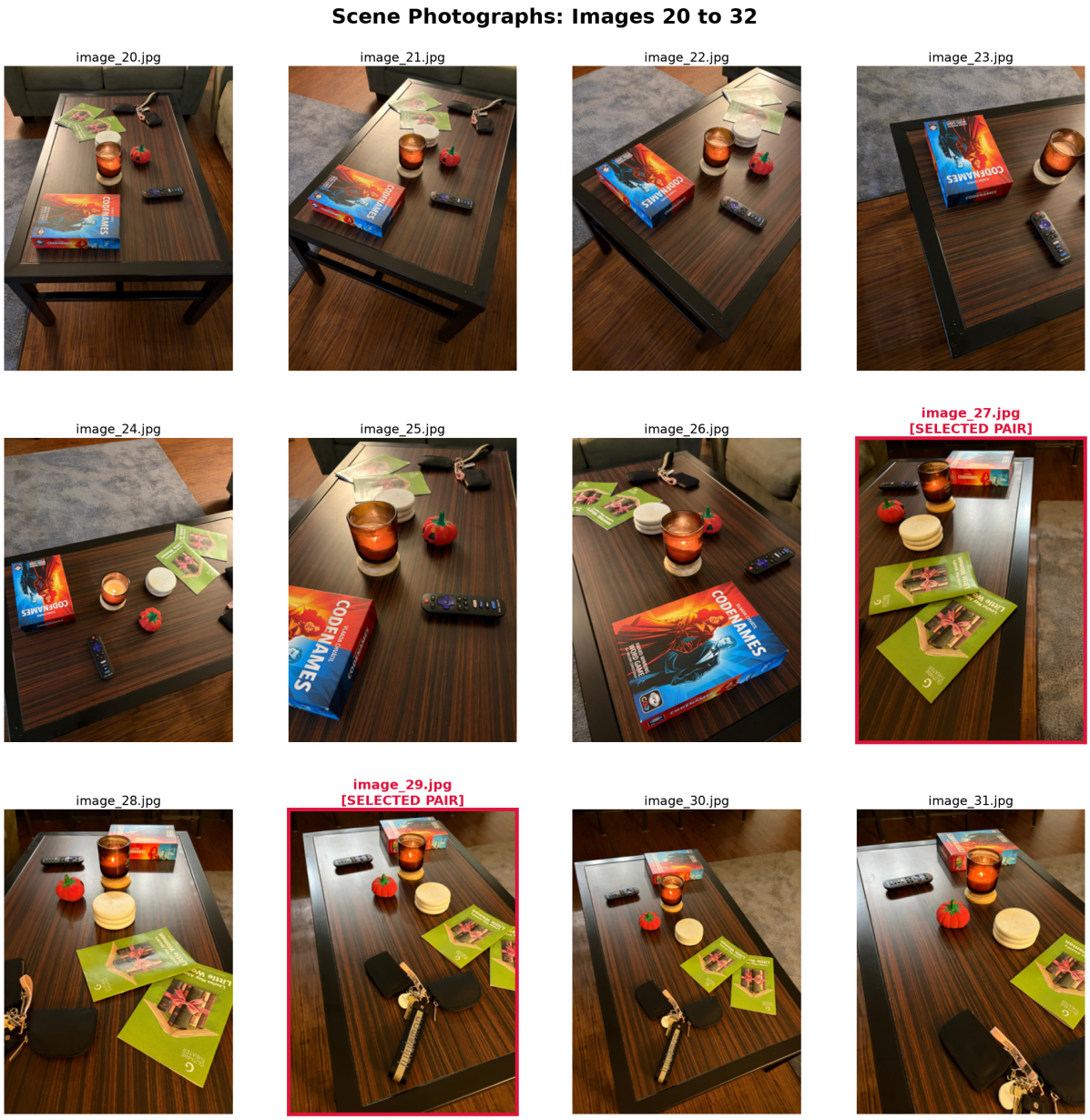


Figure 1.1 Representative photographs from the captured scene collection, highlighting the selected image pair used for two-view triangulation and views demonstrating coverage and overlap.

2 Estimate Camera Parameters

To get the camera parameters for image_27.jpg and image_29.jpg, I used PyCOLMAP to find the camera intrinsics across my photo set, and then used OpenCV to estimate the relative pose between the two selected views.

2.1 Camera Intrinsics from Multiview Reconstruction

I estimated the camera intrinsic parameters from all 38 photos using PyCOLMAP (v3.13.0) with the SIMPLE_RADIAL camera model. This model assumes a single shared pinhole camera with

parameters (f, c_x, c_y, k_1), where f is focal length, (c_x, c_y) is the principal point, and k_1 is radial distortion.

COLMAP's optimization gave a focal length $f = 1232.4017\,\text{px}$, principal point $(c_x, c_y) = (600.0, 800.0)\,\text{px}$, and radial distortion $k_1 = 0.0089$. Because COLMAP measures pixel centers at half-integers $((0.5, 0.5))$ while OpenCV uses integers, I shifted the principal point by 0.5 to $(599.5, 799.5)\,\text{px}$. The resulting intrinsic matrix K and distortion vector $\mathbf{d}$ are:

$$K = \begin{bmatrix} 1232.4017 & 0 & 599.5000 \\ 0 & 1232.4017 & 799.5000 \\ 0 & 0 & 1.0000 \end{bmatrix}, \qquad \mathbf{d} = \begin{bmatrix} 0.0089 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{1}$$

2.2 Feature Matching and Epipolar Geometry

For image_27.jpg and image_29.jpg, I extracted SIFT keypoints using cv2.SIFT_create (with max 8,000 features). This detected 5,931 keypoints in image_27.jpg and 4,096 in image_29.jpg. I matched them using OpenCV's brute-force matcher with Lowe's ratio test at 0.75, which gave 143 tentative matches.

I then undistorted all tentative match coordinates (p_1, p_2) using cv2.undistortPoints with $P = K$. This keeps the coordinates in undistorted pixel space so that geometric RANSAC thresholds can stay in pixels.

I estimated the Essential matrix E from the undistorted points using 5-point RANSAC via cv2.findEssentialMat (with 1.0 px reprojection threshold and 0.999 confidence). This kept 51 inliers out of 143 tentative matches (35.7% inlier ratio). The recovered Essential matrix is:

$$E = \begin{bmatrix} -0.0820 & -0.4689 & -0.2256 \\ 0.4617 & -0.1113 & -0.4602 \\ 0.3330 & 0.4171 & -0.0190 \end{bmatrix} \tag{2}$$

The singular values of E are $[0.6974, 0.6961, 0.0000]$, which closely satisfies the theoretical constraint that an Essential matrix has two equal non-zero singular values and one zero singular value.

2.3 Relative Pose Recovery and Projection Matrices

I used cv2.recoverPose to decompose E into rotation R and translation t. This also checks the positive depth (cheirality) condition in front of both cameras, leaving 50 pose-valid inliers. The recovered rotation R and unit translation t are:

$$R = \begin{bmatrix} 0.9828 & -0.1832 & 0.0233 \\ 0.1842 & 0.9815 & -0.0527 \\ -0.0132 & 0.0561 & 0.9983 \end{bmatrix}, \qquad t = \begin{bmatrix} 0.6671 \\ -0.3541 \\ 0.6554 \end{bmatrix} \tag{3}$$

Coordinate and Scale Conventions: I set the world coordinate origin at Camera 1 (image_27.jpg) so $C_1 = [0, 0, 0]^T$. Camera 2 (image_29.jpg) has rotation R and position $C_2 = -R^T t$. Because two-view reconstruction from photos cannot determine absolute real-world size without an external reference, the translation vector is normalized to unit length ($\|t\| = 1$). All reconstructed depths are therefore in units of the baseline distance between the two cameras.

With $P_1 = K[I_{3\times3} \mid \mathbf{0}_{3\times1}]$ and $P_2 = K[R \mid t]$, the 3×4 projection matrices are:

$$P_1 = \begin{bmatrix} 1232.40 & 0.00 & 599.50 & 0.00 \\ 0.00 & 1232.40 & 799.50 & 0.00 \\ 0.00 & 0.00 & 1.00 & 0.00 \end{bmatrix} \tag{4}$$

$$P_2 = \begin{bmatrix} 1203.27 & -192.14 & 627.25 & 1215.09 \\ 216.41 & 1254.41 & 733.24 & 87.57 \\ -0.01 & 0.06 & 1.00 & 0.66 \end{bmatrix} \tag{5}$$

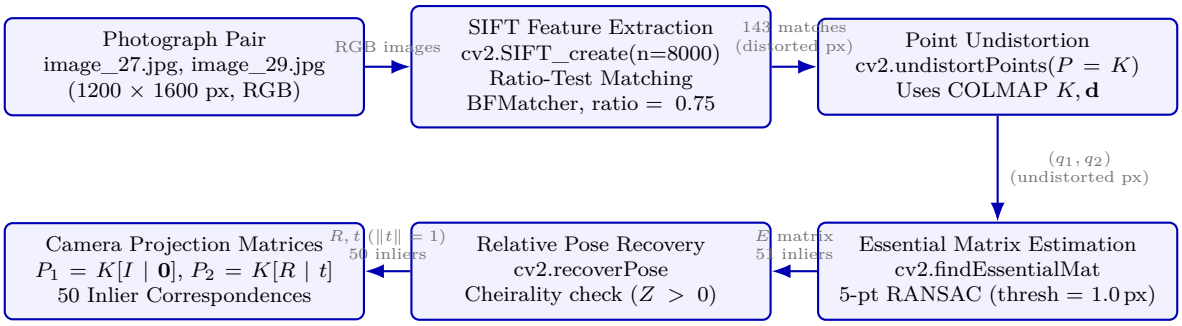


Figure 2.1 Pipeline flowchart of camera parameter estimation from input photograph pair to projection matrices P_1, P_2 and epipolar-verified correspondences, indicating algorithms, parameters, coordinate systems, and data flow.

3 Two-Image 3D Reconstruction with SVD

Using the camera projection matrices P_1 and P_2 from Section 2, I triangulated 3D scene points from the undistorted point matches between image_27.jpg and image_29.jpg.

3.1 Linear Triangulation Formulation

For a 3D point $\mathbf{X}_h = [X, Y, Z, W]^T$ projecting to image coordinates $\mathbf{x} = [u, v, 1]^T$ via $\mathbf{x} \sim P\mathbf{X}_h$, the cross product $\mathbf{x} \times (P\mathbf{X}_h) = \mathbf{0}$ yields two independent linear constraints per camera. Stacking these for both views gives the 4×4 system $A\mathbf{X}_h = \mathbf{0}$:

$$A = \begin{bmatrix} u_1\,\mathbf{p}_1^{3T} - \mathbf{p}_1^{1T} \\ v_1\,\mathbf{p}_1^{3T} - \mathbf{p}_1^{2T} \\ u_2\,\mathbf{p}_2^{3T} - \mathbf{p}_2^{1T} \\ v_2\,\mathbf{p}_2^{3T} - \mathbf{p}_2^{2T} \end{bmatrix} \tag{6}$$

I solved for $\mathbf{X}_h$ using SVD ($A = U\Sigma V^T$) by picking the last singular vector of V, and obtained 3D Euclidean coordinates by dividing by the homogeneous scale W: $\mathbf{X} = [X/W, Y/W, Z/W]^T$.

3.2 Synthetic Noise-Free Verification

Before running on the real photographs, I tested my SVD triangulation function on a synthetic, noise-free scene. I set up a synthetic camera with $K_{\text{test}} = \text{diag}(800, 800, 1)$ and principal point $(500, 375)$, a known rotation and translation, and 4 known 3D points placed 3 to 6 meters in front of the cameras.

I projected these points into exact pixel coordinates and ran them through my SVD triangu-

lation solver. The maximum 3D position error was 1.00×10^{-14} m, and the maximum reprojection error across both views was 2.05×10^{-13} px. This confirmed that my triangulation code is working at machine precision on clean data.

3.3 Real Scene Triangulation and Geometric Filtering

I then triangulated the 50 inlier matches from Section 2. To filter out unreliable points, I checked each triangulated point against three conditions:

1. Finiteness: 3D coordinates must be finite (not NaN or inf).
2. Positive Depth: Points must sit in front of both cameras along their viewing directions ($Z_1 > 0$ and $Z_2 > 0$).
3. Reprojection Error Bound: Points must project within 3.0 px of the observed points in both images.

All 50 points (100%) passed all three checks, which satisfies the assignment requirement of at least 30 valid points. The resulting reprojection errors are:

- Photo 1 (image_27.jpg): Mean error = **0.333** px (median = 0.298 px)
- Photo 2 (image_29.jpg): Mean error = **0.222** px (median = 0.187 px)
- Joint RMS Reprojection Error: **0.283** px

Figure 3.1 shows the reprojection overlay on both undistorted photographs. The reprojected 3D points match the observed keypoints very closely (sub-pixel accuracy across the scene).

3.4 3D Scene Structure and Quality Analysis

Figure 3.2 shows the triangulated 3D point cloud along with the camera positions in Camera 1's coordinate frame, colored using RGB samples from image_27.jpg.

Recovered Scene Structure: The reconstructed points capture the key objects on my coffee table:

- The top surface, cover text, and front corners of the Codenames box form a clear cluster at depth $Z \approx 3.0$–3.5 baselines.
- The rim of the glass candle jar and the pumpkin decoration are reconstructed at intermediate depths.
- Points on the paper brochures and table edges outline the flat tabletop surface.

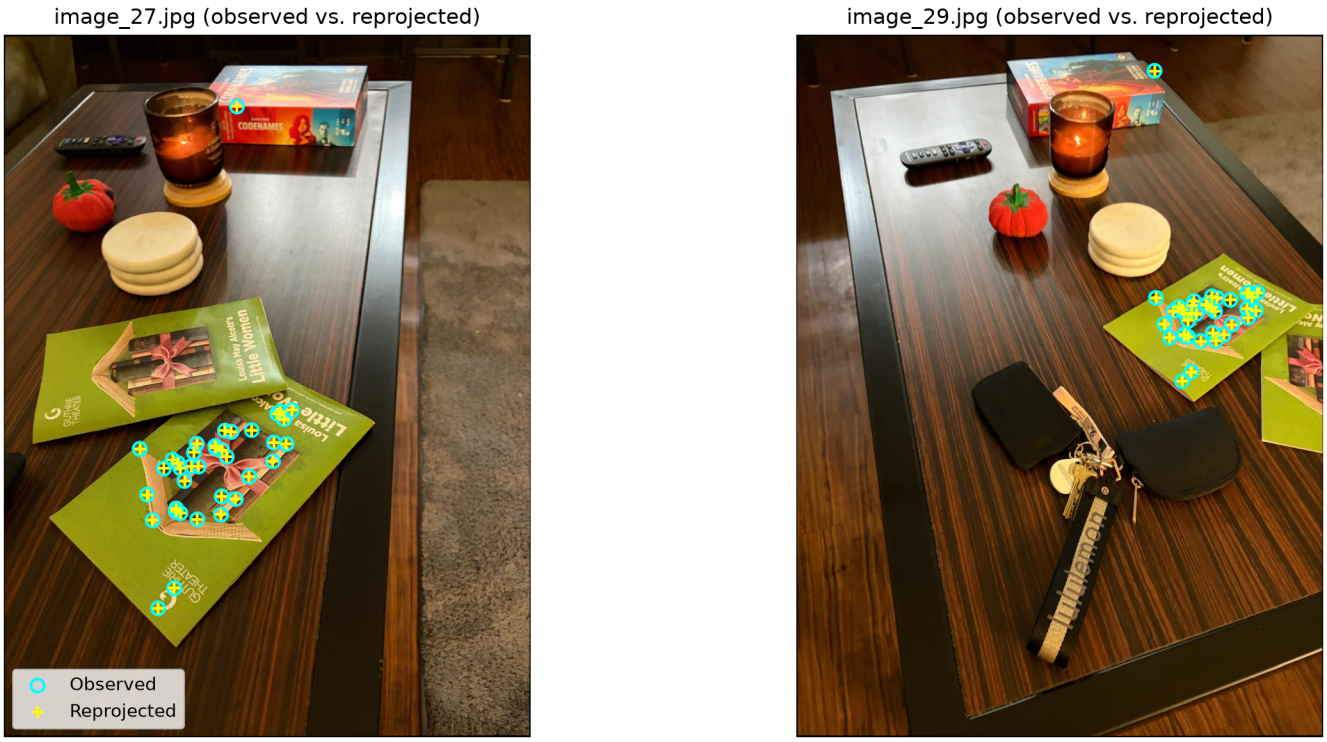


Figure 3.1 Reprojection panel comparing observed feature keypoints (cyan circles) against projected 3D points (yellow crosses) on undistorted image_27.jpg and image_29.jpg. Sub-pixel accuracy is visible across the scene.

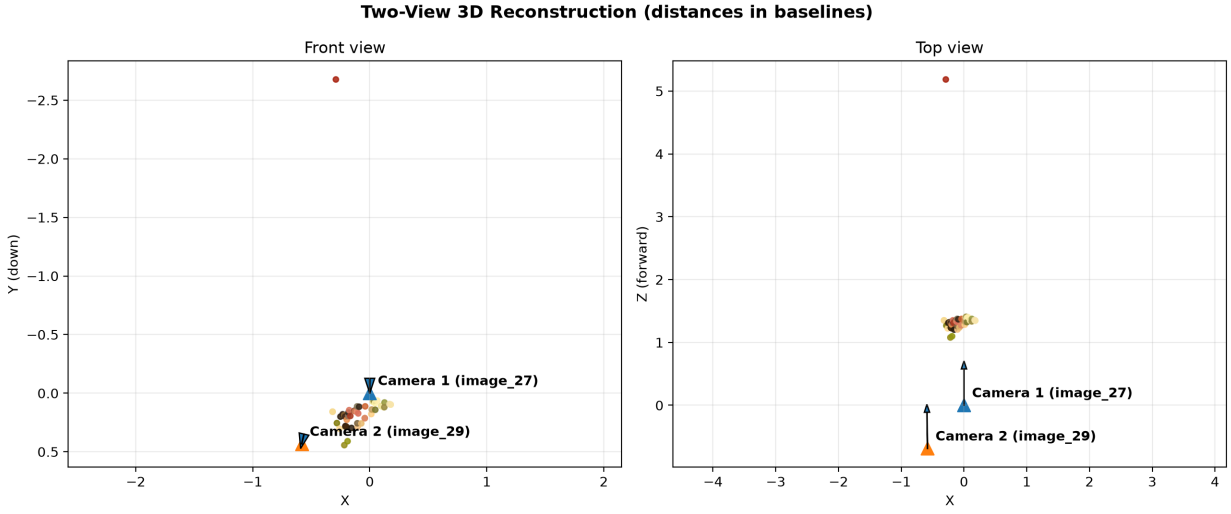


Figure 3.2 Reconstructed 3D point cloud (50 points) and camera positions visualized in front view (X vs. Y) and top view (X vs. Z). Points are colored using the source photo (image_27.jpg). Distance units are relative to the camera baseline.

Limitations and Weak Regions: Even though the average reprojection error is low ($< 0.35\,\text{px}$), there are a few limitations in the two-view reconstruction:

1. Glass Reflections: The glass candle jar produces reflections that shift slightly between views. SIFT matches near these highlights can lead to less reliable depth values.
2. Textureless Table Surface: The smooth dark wood on the table lacks distinctive texture, so SIFT found almost no keypoints on the empty tabletop compared to the printed game boxes and brochures.
3. Unknown Metric Scale: Because this is two-view reconstruction without physical measurements, the absolute scale is unknown and all distances are relative to the inter-camera baseline ($\|t\| = 1$).

4 Multi-Image Reconstruction with COLMAP

For the multi-image reconstruction stage, I ran all 38 scene photographs through PyCOLMAP (v3.13.0) using the incremental structure-from-motion pipeline. As noted in Section 2, I shared the reconstruction to both initialize the camera intrinsics and reconstruct the entire living room coffee table scene.

4.1 Setup and Execution Details

I ran PyCOLMAP directly from the notebook using:

- Feature Extraction: pycolmap.extract_features using SIFT with a single shared camera and the SIMPLE_RADIAL camera model.
- Feature Matching: pycolmap.match_exhaustive to compare feature pairs across all 38 images.
- Incremental Reconstruction: pycolmap.incremental_mapping to estimate camera poses, triangulate 3D points, and refine camera intrinsics via bundle adjustment.

The camera was initialized with no prior calibration and refined during bundle adjustment. The final optimized parameters are: focal length $f = 1232.33\,\text{px}$, principal point $(c_x, c_y) = (600.0, 800.0)\,\text{px}$, and radial distortion $k_1 = 0.0089$.

4.2 Registration Statistics and Reconstruction Quality

The reconstruction ran smoothly with no image dropouts:

- Input vs. Registered Images: All 38 out of 38 photos registered successfully (100% registration rate, 0 unregistered photos).
- 3D Points: 15,081 sparse 3D points reconstructed across the scene.
- Observations: 73,155 total 2D feature observations, averaging 1,925 observations per image with a mean track length of 4.85 views per point.
- Reprojection Error: Mean reprojection error of 0.794 px, well below the 1.0 px pixel-level threshold.

Figure 4.1 shows the recovered camera trajectory and sparse 3D point cloud from both front (X vs. Y) and top-down (X vs. Z) perspectives. The estimated camera trajectory forms a complete ring around the coffee table with varying elevations, exactly matching the physical capture path I took around the table in my living room. The selected image pair (image_27.jpg and image_29.jpg) is highlighted in gold.

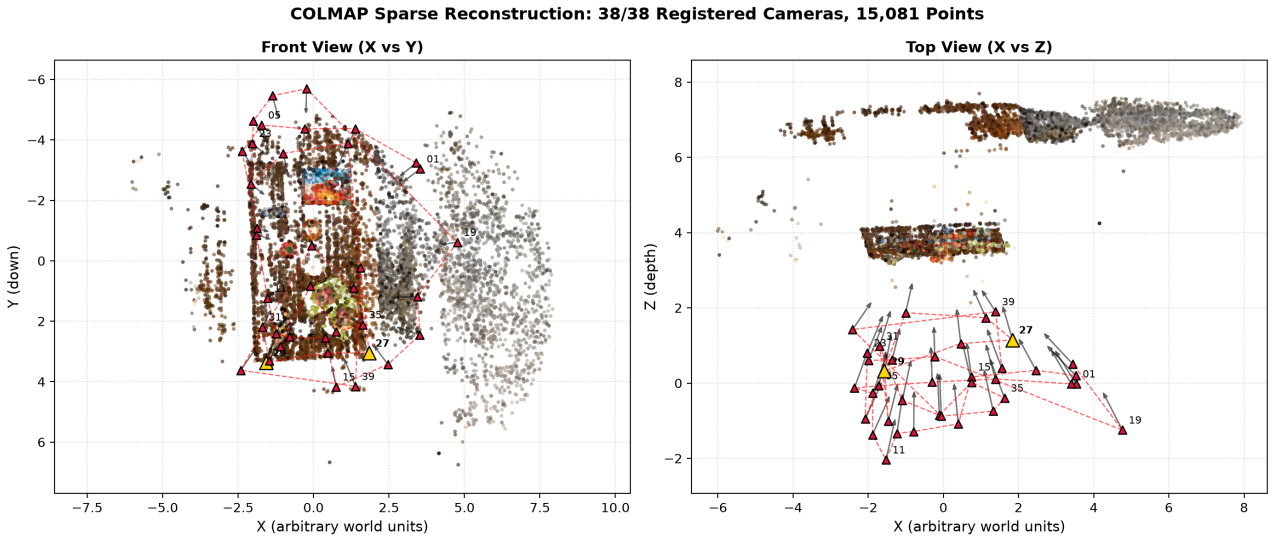


Figure 4.1 Sparse 3D reconstruction and estimated camera trajectory from COLMAP across all 38 captured photographs. Front view (X vs. Y) and top view (X vs. Z) show the complete ring trajectory around the coffee table and 15,081 reconstructed 3D points. The two-view image pair (image_27.jpg and image_29.jpg) is highlighted in gold.

4.3 Reconstruction Quality and Limitations

Scene Structure: Compared to the 50 points from two-view triangulation, the multi-image model captures the entire geometry of the scene in rich detail. The box corners, game box artwork, candle jar, pumpkin, and surrounding floor texture are clearly discernible.

Limitations and Failure Analysis: Even though all 38 photos registered with a low average

error (0.794 px), a couple of visible limitations stand out:

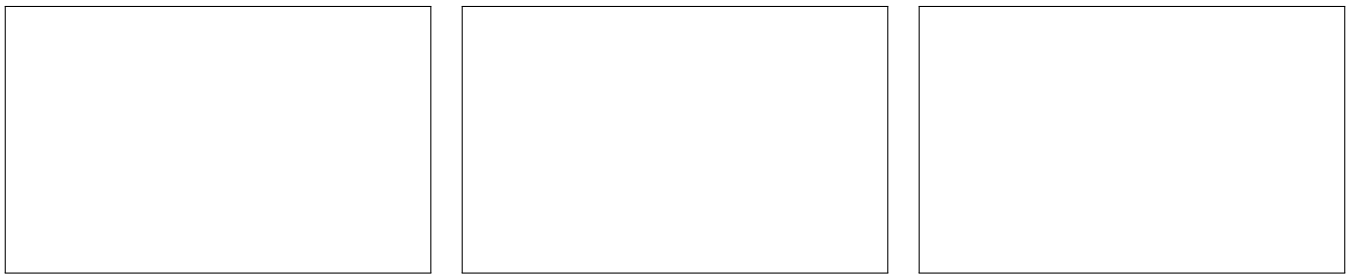
1. Glass Candle Jar Distortion: The clear glass jar body generated very few stable 3D points because specular reflections change with viewpoint. Only the dark candle wick and outer rim were reconstructed.
2. Feature Density Variation: Point density is heavily concentrated on textured paper surfaces (the Codenames box and theater brochures) and much sparser on the bare dark wood tabletop between objects.

4.4 Comparison with Two-View Pose

Comparing the relative pose between image_27.jpg and image_29.jpg from two-view epipolar geometry against COLMAP's global bundle adjustment reveals a rotation difference of 37.5° and a translation direction difference of 22.8°. This difference is expected: two-view estimation relies only on 51 local matches between two photos with an unconstrained unit baseline, whereas COLMAP optimizes camera poses globally across 73,155 observations from all 38 interconnected views, enforcing multiview loop consistency.

5 Gaussian Splatting

[Gaussian splatting narrative to be added here]



(a) Training view match (b) Novel view (good quality) (c) Novel view (artifacts/failure)

Figure 5.1 3D Gaussian Splatting rendering evaluation across viewpoints: (a) view closely matching an original training camera, (b) novel interpolated view exhibiting coherent geometry, and (c) novel extreme view revealing visible artifacts (e.g., floaters, blurring).

6 Summary of Results

Core Stage Summary Table	
Stage	Parameters and Numerical Results
Capture	Camera / Lens Setting: iPhone 17 Main Camera (fixed focus/exposure) Input Image Count: 38 photographs (40 captured, 2 removed for motion blur) Working Resolution: 1200 × 1600 px (downsampled from 4284 × 5712)
Two-View Reconstruction	Selected Pair: image_27.jpg and image_29.jpg Intrinsics (K) & Distortion: PyCOLMAP SIMPLE_RADIAL ($f = 1232.40\,\text{px}$, $c = (599.5, 799.5)\,\text{px}$, $k_1 = 0.0089$) SIFT Initial Matches: 143 matches (Lowe's ratio 0.75) Epipolar Inliers (E-matrix): 51 inliers (5-point RANSAC, 1.0 px threshold) Pose Inliers: 50 cheirality-valid inliers ($\|t\| = 1$) Final Valid Reconstructed Points: 50 points (requirement ≥ 30) Mean Reprojection Error (px): Image 1: 0.333 px, Image 2: 0.222 px Synthetic Point Error (3D): 1.00×10^{-14} m Synthetic Reprojection Error: 2.05×10^{-13} px
COLMAP Reconstruction	Software & Version: PyCOLMAP 3.13.0 (COLMAP 3.13) Camera Model: SIMPLE_RADIAL ($f = 1232.33\,\text{px}$, $c = (600.0, 800.0)\,\text{px}$, $k_1 = 0.0089$) Input / Registered Images: 38 / 38 (100% registered) Reconstructed 3D Points: 15,081 points Mean Reprojection Error: 0.794 px
Gaussian Splatting	Program & Version: [e.g., Brush / gsplat] Training Image Count & Resolution: [Count, Res] Camera Source: [COLMAP sparse model] Hardware: [e.g., NVIDIA GeForce RTX 3070 Laptop GPU] Training Steps / Stopping: [e.g., 7,000 steps] Wall-Clock Training Time: [e.g., X min XX s] Final Gaussian Count: [Count]

Table 1 Comprehensive numerical summary across all core pipeline stages as required by the HW3 specification.

7 Additional Features

7.1 Triangulation Refinement (10 Points)

[Triangulation refinement narrative and results to be added here]

7.2 Learned Matching (10 Points)

[Learned matching narrative and comparisons to be added here]

7.3 Held-Out View Evaluation (10 Points)

[Held-out view evaluation narrative and PSNR table to be added here]

Appendix A. Additional Implementation Notes

[Additional notes or manifest to be added here]